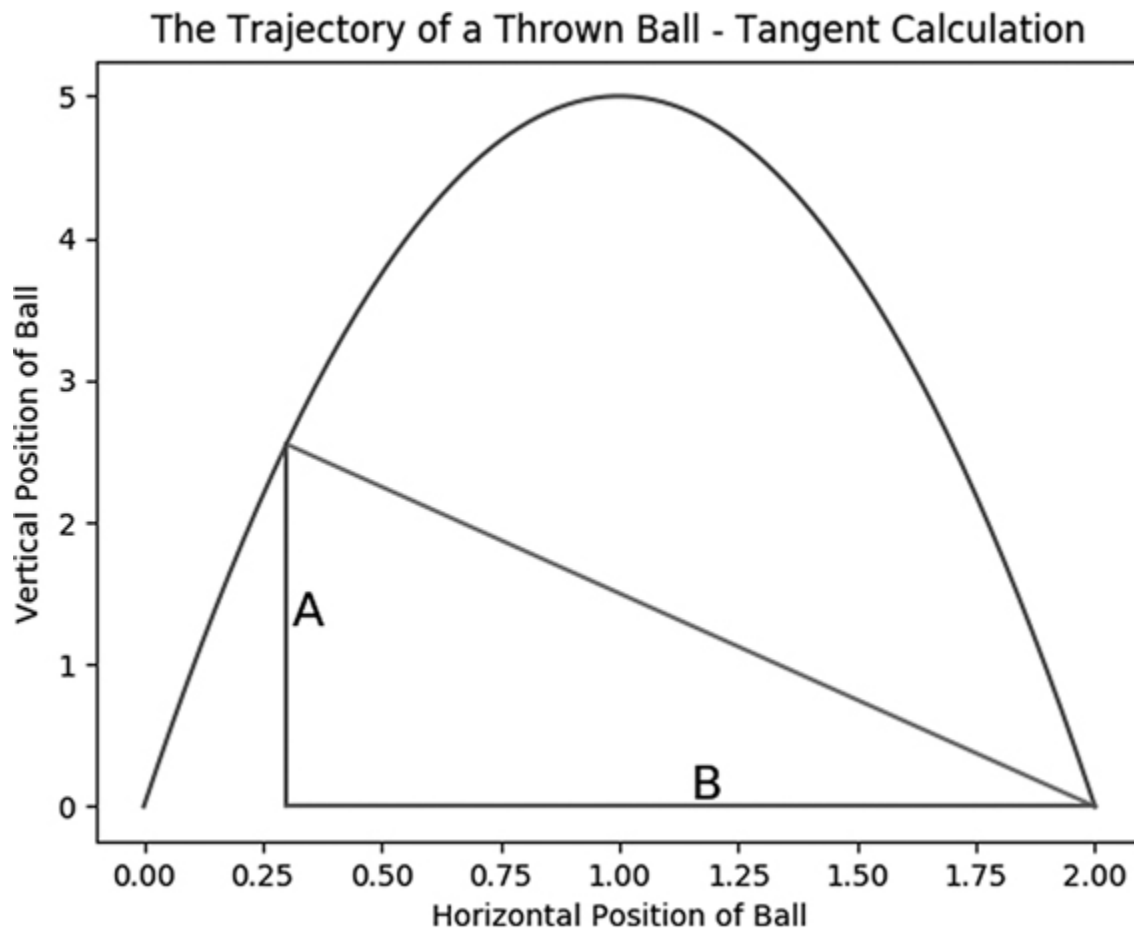


```

ys6 = [0,0]
plt.title('The Trajectory of a Thrown Ball - Tangent Calculation')
plt.xlabel('Horizontal Position of Ball')
plt.ylabel('Vertical Position of Ball')
plt.plot(xs,ys,xs4,ys4,xs5,ys5,xs6,ys6)
plt.text(0.31,ball_trajectory(0.3)/2,'A',fontsize = 16)
plt.text((0.3 + 2)/2,0.05,'B',fontsize = 16)
plt.show()

```

The resulting plot is shown in [Figure 1-3](#).



[Figure 1-3](#): The trajectory of a hypothetical thrown ball, with a line segment representing the outfielder looking at the ball as it travels, and line segments A and B showing the lengths whose ratio constitutes the tangent we are interested in

We calculate the tangent by taking the ratio of the length of the side labeled A and the length of the side labeled B. The equation for the height A will be $10x - 5x^2$, while the equation for the length of B will be $2 - x$. So the